

API Data Collection

Lesson 08 · ISBA 4715

Agenda

	Part	What
1	API Concepts	What APIs are and why they matter (<i>slides</i>)
2	Weather API by Hand	Your first API call (<i>live code</i>)
3	Loops + Bulk Collection	Getting enough data (<i>AI-assisted</i>)
4	Your Project Connection	Finding APIs for your domain

What apps do you use?

Personal or work. Submit your top two on Teams.



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APIs Are Everywhere

When you check the weather on your phone → **API call**

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When you check the weather on your phone → **API call**

When Uber charges your card → **API call**

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The apps you use are wrappers around other services, connected by APIs.



API = Restaurant

Role	What They Do
You	Place an order



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Waiter	Carry it to the kitchen



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You never enter the kitchen.

You don't need to know how it works.

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"Give me the current weather for Los Angeles"

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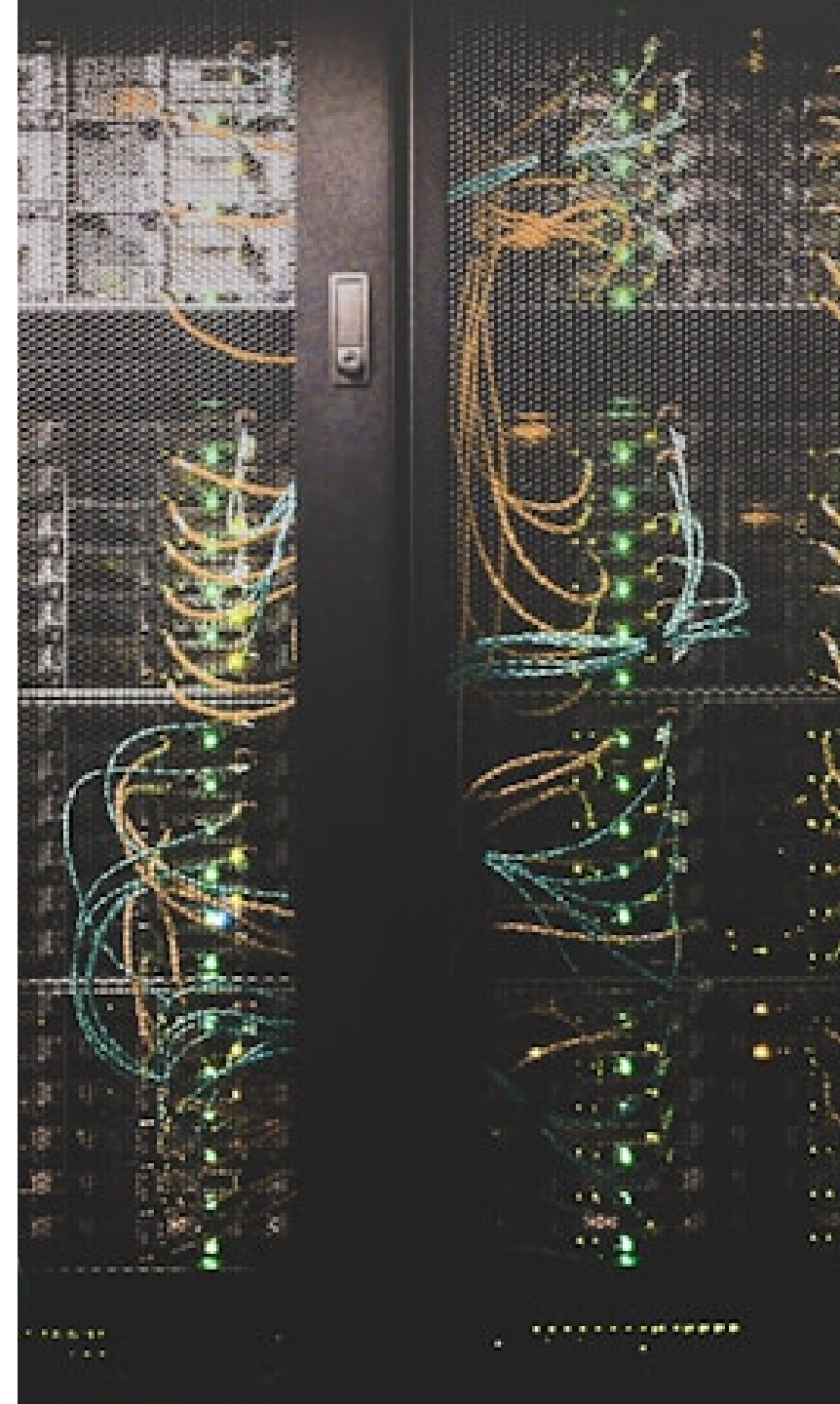
You get back a response:

```
{ "temp_f": 72.0, "condition": "Sunny" }
```

REST: Request + Response

Two actions:

- **GET** = "Give me data"
- **POST** = "Create something"



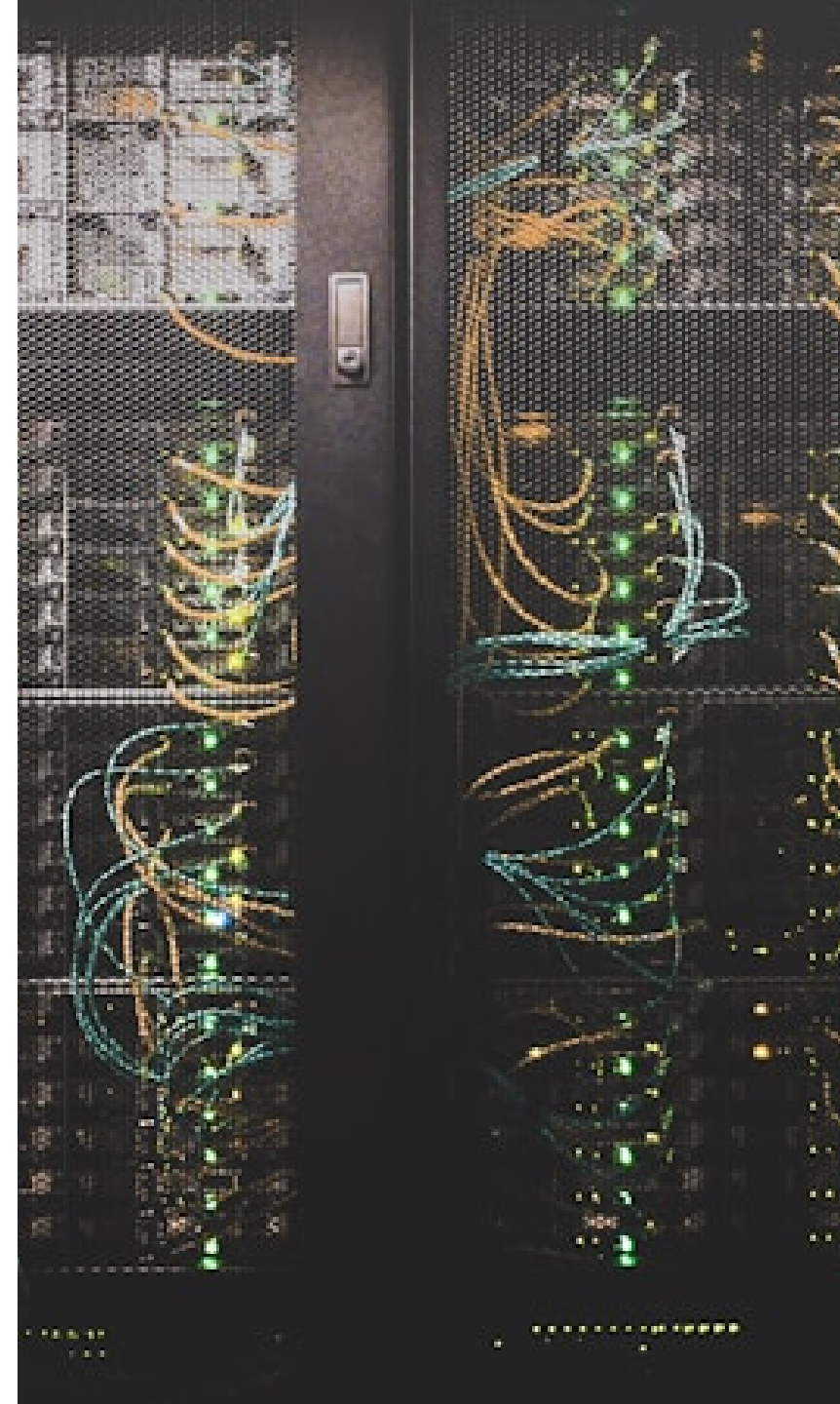
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Three parts to a request:

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- **Parameters** — what you want
- **API key** — who you are



REST: Request + Response

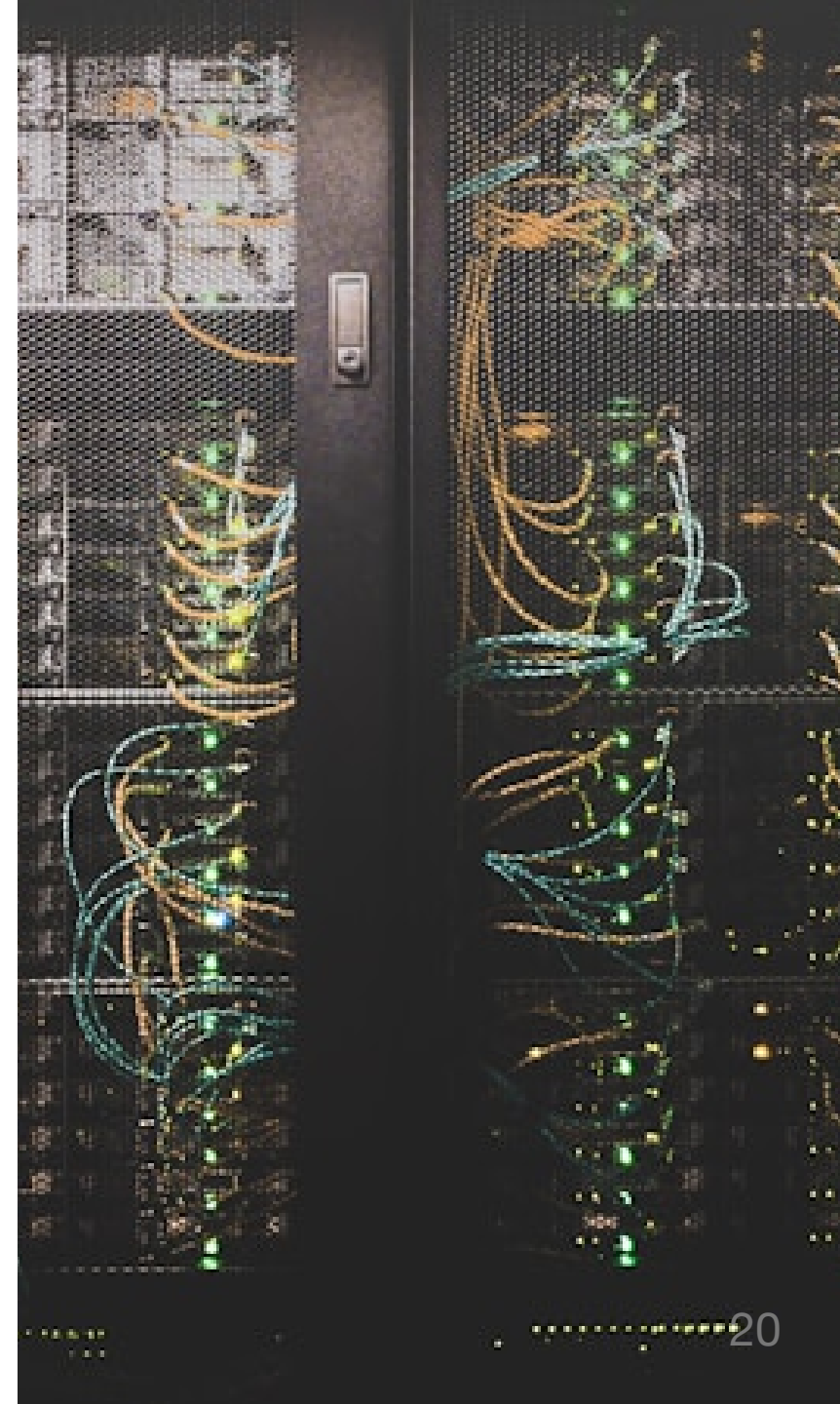
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Response comes back as **JSON**.



JSON: The Language of APIs

APIs return data in JSON — a structured text format:

```
{  
  "location": {  
    "name": "Los Angeles",  
    "region": "California"  
  },  
  "current": {  
    "temp_f": 72.0,  
    "condition": {  
      "text": "Sunny"  
    }  
  }  
}
```

{ } = object [] = list Data is **nested** — objects inside objects

API Keys

- APIs require a **key** to identify you
- Free tier = limited calls per day
- **Never commit keys to GitHub**

Store keys in `.env` files.

Think of the key as a library card:
it tracks usage and enforces limits.



Build vs. Buy

Need	Don't Do This	Do This
Weather data	Build a weather station	Call the Weather API

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Financial data	Download CSVs manually	Call a market data API

APIs let you access data without building the source yourself.

One API Call = One Data Point

```
GET /current.json?q=90045
```

gives you weather for **one** zip code, **right now**

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Your project needs hundreds of rows. How?

Loops = Datasets

20 locations → Weather for 20 cities → **20 rows**

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One API call = one row. **A loop = a dataset.**

The Pattern

request → parse → loop → DataFrame →
save

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Where does it go?

CSV, SQL database, JSON file, Snowflake, another API, a dashboard

This works for any API and any destination. Once you know it, switching is mostly reading documentation.

Your Project Needs an API

- Milestone 01 requires an API source



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Next up: code it together

